

SQL PROJECT 1

DODS @ NDMIT

Data Analysis Report & Business Problems

A structured analytical walkthrough of Anita's Veg Café — how each business problem, mapped the data model, and framed the questions that SQL will answer to drive real decisions.

ROLE	CLIENT	TOOL	PROBLEMS
Data Analyst	Anita's Veg Café	PostgreSQL	11 Business Questions

MY ANALYTICAL APPROACH

How I Framed & Solved This

Before writing a single line of SQL, I took time to understand the business context — what decisions Anita is trying to make, what data she has, and how the tables relate to each other. Here is the structured methodology I followed.



Understand the Business First

I mapped each question back to a real business decision. Revenue questions inform pricing strategy. Popularity queries drive inventory planning. Loyalty queries measure program ROI. Churn detection enables proactive retention. This framing ensured every query had a purpose beyond just returning a result.



Audit the Data Model

I examined all three tables — **sales**, **menu**, and **members** — and mapped their relationships. Sales is the central fact table; menu is a dimension/reference table; members is an event table. This distinction shaped every JOIN and filter condition.

1

2



Choose the Right SQL Technique

I deliberately chose between GROUP BY aggregations and window functions based on what each question needed. Rankings and first/last event detection called for DENSE_RANK or RANK. Simple totals and counts used GROUP BY. Multi-step logic used CTEs for readability and maintainability.



Translate Results to Insights

A query result is only half the answer. For each problem, I noted what the output means in plain business language — so Anita, or any non-technical stakeholder, can read the findings and make confident decisions without needing to understand the SQL underneath.

DATA MODEL

Entity Relationship Overview

The schema `anitas_veg_cafe` contains three tables. **sales** is the central fact table. It links to **menu** via `product_id` and to **members** via `customer_id`. Not every customer in sales has a loyalty record — the members join is optional.

SALES	
◦ customer_id	VARCHAR(1)
order_date	DATE
◦ product_id	INTEGER
product_id →	
MENU	
◦ product_id	INTEGER
product_name	VARCHAR(5)
price	INTEGER
MEMBERS	
◦ customer_id	VARCHAR(1)
join_date	TIMESTAMP

SALES → MENU Many-to-one. Many transactions reference one menu item via product_id. This JOIN retrieves product names and prices for every order.	SALES → MEMBERS Many-to-one, nullable. Not every customer in sales has a members record. Used with date comparisons to isolate pre/post loyalty behaviour.	KEY DESIGN NOTE Rohan appears in sales but has no members record. Loyalty-related queries use INNER JOIN on members, which naturally excludes non-members — an important boundary condition.
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BUSINESS PROBLEMS

11 Questions to Answer

Each card below represents a real analytical need at the café — reframed as an industry-style business problem. I've described how I approached it, what SQL technique is appropriate, and what business value the answer unlocks. Click any card to expand.

Q 01

Customer Lifetime Revenue

What is the cumulative spend of each customer at the café?

REVENUE ANALYSIS

ANALYTICAL APPROACH

Before building any loyalty strategy or running promotions, Anita needs a clear baseline of who her most valuable customers are by **total revenue contributed**. This is the single most foundational metric in customer analytics — it separates high-value regulars from casual visitors even if both come in the same number of times.

To solve this, I need to join the **sales** table with **menu** to bring in the price for each transaction, then aggregate total spend grouped by customer.

SQL TECHNIQUES USED

JOIN (sales × menu)

SUM() OVER(PARTITION BY)

DISTINCT

BUSINESS VALUE

☞ Why This Matters

This tells Anita which customers are driving the most revenue. A customer who visits frequently but always orders the cheapest item may be less valuable than a less frequent high-spender. This metric directly informs reward tier design and personalised offers — it's the entry point to any customer segmentation model.

Q 02

Customer Visit Frequency

How many unique visit days has each customer made to the café?

▼

Q 03

First Purchase Item per Customer

What dish created the first impression for each customer?

▼

Q 04

Overall Best-Selling Dish

Which single menu item drives the highest order volume across all customers?

▼

Q 05

Top Dish per Customer

What is each customer's personal go-to dish at the café?

▼

Q 06

First Post-Membership Purchase

What was the first dish each loyalty member ordered after enrolling in the program?

▼

Q 07

Last Pre-Membership Purchase

What was the final dish each customer ordered before committing to the loyalty program?

▼

Q 08

Pre-Membership Spend & Volume

How much did loyalty members spend and how many items did they purchase before joining?

▼

Q 09

Loyalty Points Balance — Standard Rules

Based on the standard points structure, how many points has each customer accumulated?

▼

Q 10

January Points Total — With Bonus Week

Accounting for the first loyalty week double-points bonus, what are Aarav and Meera's total points by end of January?



Q 11

Churn Risk Detection

Which customers have not placed an order in the last 30 days and may be disengaging?



Anita's Veg Café

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PostgreSQL · Schema: anitas_veg_cafe